

K1296 — Casey’s sphere-submanifold rescue of Obata: the DE relaxation lives on the Shilov BOUNDARY, which IS spheres ($S^4 \times S^1 / Z_2$). $\lambda_1 = E$ reopened as a CANDIDATE (not re-banked, not dead).

Keeper, 2026-08-08 (last turn before refresh). K1294 retracted “ $\lambda_1 = \text{Einstein constant by Obata}$ ” because the BULK compact dual Q^5 is a Hermitian symmetric space, not a sphere ($E < \lambda_1$ there). Casey’s point: BST HAS spheres — “a sphere tiled by circles” — so there may be a submanifold where Obata’s equality DOES hold. This has real traction and reopens the tie as a candidate. Grep-before-reopening honored: this explicitly references the K1294 retraction; it is a candidate reopening, NOT a re-bank. Cal audits.

Why the rescue has traction (it’s not a stretch)

The dark-energy relaxation is a boundary phenomenon: the records/commitments live on the Shilov boundary, and the bleed is the boundary relaxing toward the zero-mode (Elie built the modes as “SO(5) harmonics on the Shilov boundary $S^4 \times S^1 / Z_2$ ”). The Shilov boundary IS built from spheres — $S^4 \times S^1 \pmod{Z_2}$. So even though the *bulk* Q^5 (where Obata fails) is a Hermitian symmetric space, the relaxation the DE rate reads lives on the sphere-boundary, where Obata CAN apply. Casey’s “sphere tiled by circles” is exactly this: an S^4 (or S^5) carrying the S^1 commitment-circle (a Hopf-type fibration in the corpus, K1238). The $\lambda_1 = E$ identity was retracted for the bulk; on the boundary sphere it is a live candidate.

What the rescue must show (the computation — target-innocent)

1. Which sphere carries the relaxation? S^4 (the Shilov factor), the total $S^4 \times S^1 / Z_2$, or a round S^m . On a round S^m , $\lambda_k = k(k+m-1)$; Obata’s equality $\lambda_1 = (\text{Einstein constant})$ holds *on the sphere*.
2. Reconcile the corpus formula $\lambda_k = k(k+5)$. $k(k+5) = k(k+(m-1)) \iff m = 6$ (round S^6), OR it is the Q^5 own-formula. This is the crux: if $k(k+5)$ is genuinely a round-sphere spectrum (which S^m ?), Obata applies to THAT sphere and $\lambda_1 = E$ is rescued there; if it’s the Hermitian Q^5 formula, the retraction stands. Verify against the actual spectrum — do NOT assume (that was the original error).
3. Normalization: the Einstein constant and λ_1 must be read in the same metric normalization on the sphere for the equality to mean anything.

Tier (held — do not grab the rescue too fast)

- RETRACTED still stands for the BULK Q^5 ($E < \lambda_1$; Obata is sphere-only) — K1294.
- REOPENED as a CANDIDATE for the boundary sphere — plausible because the DE relaxation is a Shilov-boundary phenomenon and the Shilov boundary is spheres. NOT re-banked — grabbing the rescue on “there’s a sphere somewhere” would be the same over-claim in reverse. It banks only when (1)–(3) are shown, target-innocent.
- Unaffected regardless: the DE SIGN (Bernstein) banks; the discriminator; the #79 reduction (rests on $c_{IR} = C_2$, not Obata).

Next

Compute: identify the sphere the relaxation lives on, verify whether $\lambda_k = k(k+5)$ is that sphere’s spectrum, and whether Obata gives $\lambda_1 = E$ there with consistent normalization. If yes, the $a_0 \leftrightarrow a_1$ tie (“DE relaxes at the Einstein-constant rate”) is rescued on the boundary sphere — a genuinely tight result, earned. If no, the retraction is final. Either way, target-innocent, verified at source.

— Keeper, K1296, 2026-08-08. Casey’s sphere-submanifold rescue: the DE relaxation is a Shilov-BOUNDARY phenomenon, and the Shilov boundary IS spheres ($S^4 \times S^1 / Z_2$, a sphere tiled by the S^1 commitment-circle), so Obata (sphere-only) may apply THERE even though the bulk Q^5 (Hermitian symmetric) fails it. $\lambda_1 = E$ REOPENED as a candidate — NOT re-banked. Computation: which sphere, is $\lambda_k = k(k+5)$ its spectrum ($k(k+5) \stackrel{?}{=} S^6$?), Obata + normalization. References K1294 retraction (honest reopening). DE sign / discriminator / #79-reduction unaffected. Cal audits. Nothing pushed.